

MINNEAPOLIS ANOKA COUNTY AP, MN, USA

WMO: 726577

Lat: 45.150N

Lon: 93.217W

Elev: 912

StdP: 14.22

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 94974

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.8	-4.5	-17.2	2.2	-7.9	-12.8	2.8	-4.9	23.7	19.5	20.8	15.2	6.2	290	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	90.1	74.7	87.6	73.6	83.8	71.5	78.0	85.8	75.7	83.3	73.9	81.2	12.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	136.6	82.8	73.0	126.9	80.2	71.9	122.1	79.1	41.9	86.8	39.8	83.5	38.0	80.3	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	19.7	17.7	DB	-15.0	94.4	5.9	2.5	-19.2	96.2	-22.6	97.7	-25.9	99.1	-30.2	101.0
			WB	-15.5	81.2	5.5	2.5	-19.5	83.0	-22.7	84.5	-25.8	85.9	-29.8	87.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	16.0	23.3	31.0	46.5	56.6	66.6	72.1	69.5	62.9	49.3	35.4	22.6
	DBStd	21.37	12.07	11.55	11.45	9.90	8.22	7.32	6.29	5.81	8.80	9.63	10.11	11.89
	HDD50	4043	1053	746	598	176	30	0	0	0	7	133	451	849
	HDD65	7524	1518	1166	1055	560	283	67	13	23	142	496	888	1313
	CDD50	2620	0	0	8	73	235	499	685	603	393	111	13	0
	CDD65	624	0	0	0	6	22	115	233	161	78	9	0	0
	CDH74	5557	0	0	0	75	257	1025	2165	1318	657	60	0	0
	CDH80	1704	0	0	0	25	75	296	747	364	187	10	0	0
Wind	WSAvg	7.8	7.9	8.0	8.4	8.9	8.6	7.4	6.5	6.3	7.7	8.0	7.9	7.9
Precipitation	PrecAvg	33.0	0.9	0.8	1.7	3.0	4.1	4.8	4.6	4.3	3.3	2.8	1.9	1.1
	PrecMax	43.9	2.0	1.7	3.7	7.9	9.0	8.7	9.3	7.6	7.8	6.2	5.4	2.2
	PrecMin	24.5	0.1	0.2	0.5	0.8	0.2	1.9	0.9	0.4	0.4	0.7	0.0	0.2
	PrecStd	4.8	0.5	0.4	0.8	1.5	2.0	1.8	2.0	1.9	2.0	1.7	1.4	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.3	52.1	66.1	83.7	88.1	91.0	93.2	92.9	89.7	81.2	69.5	54.9
		MCWB	40.2	45.1	54.3	66.9	68.9	74.2	75.5	77.6	72.5	68.0	55.2	49.2
	2%	DB	38.8	46.0	60.9	72.8	81.3	87.6	90.4	87.9	84.4	73.4	62.7	45.8
		MCWB	35.0	40.9	53.2	56.5	63.8	72.8	75.6	75.1	70.6	61.9	55.0	41.4
	5%	DB	35.7	43.1	54.4	68.3	75.0	82.4	87.9	83.6	81.4	69.7	55.4	42.6
		MCWB	33.9	39.2	47.0	53.9	59.7	69.4	74.6	72.4	68.2	60.5	48.5	38.2
	10%	DB	32.0	39.1	47.8	63.7	72.1	80.7	84.0	81.3	77.3	64.1	50.5	37.1
		MCWB	30.9	36.2	42.9	50.9	59.2	67.7	73.1	69.8	65.7	56.3	44.6	34.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.1	45.5	58.2	67.3	71.5	78.2	81.4	81.5	75.6	70.3	61.4	49.7
		MCDB	46.2	48.8	62.3	80.5	83.1	87.5	89.4	87.7	84.9	75.2	62.6	56.7
	2%	WB	36.2	42.0	53.1	60.4	66.6	75.0	78.6	77.0	73.1	65.1	55.5	42.6
		MCDB	39.0	45.8	59.5	69.3	76.4	83.0	86.0	83.6	80.5	71.1	61.1	46.0
	5%	WB	33.5	39.5	48.5	55.3	63.5	72.7	76.6	74.3	70.9	62.0	49.7	37.8
		MCDB	35.0	43.0	53.9	65.2	71.8	80.3	84.1	81.4	79.0	66.4	56.0	41.5
	10%	WB	30.4	36.7	42.0	52.3	60.4	70.4	74.7	71.9	68.2	58.1	45.4	34.8
		MCDB	31.6	39.5	48.1	61.7	68.6	77.6	82.0	78.8	75.5	64.1	49.8	37.4
Mean Daily Temperature Range	5% DB	MDBR	14.6	15.2	16.8	20.1	19.5	18.7	18.2	18.9	19.8	18.2	14.7	13.8
		MCDBR	15.2	15.9	24.2	27.4	25.6	21.0	20.7	21.2	22.6	23.4	23.0	18.5
		MCWBR	12.5	11.8	16.5	14.9	12.8	10.2	9.6	10.8	11.3	14.0	14.9	13.8
	5% WB	MCDBR	14.8	14.8	22.7	23.8	22.5	18.3	18.2	18.6	20.2	18.9	20.8	17.0
		MCWBR	12.8	11.3	16.4	14.9	13.9	10.9	10.1	11.7	11.5	13.3	15.5	13.3
Clear-Sky Solar Irradiance	taub	0.268	0.280	0.319	0.379	0.409	0.399	0.395	0.404	0.375	0.321	0.291	0.268	
	taud	2.410	2.361	2.376	2.286	2.250	2.328	2.359	2.337	2.385	2.529	2.514	2.433	
	Ebn at Noon	272	289	290	279	273	276	275	268	267	271	262	259	
	Edh at Noon	23	29	33	40	42	39	38	37	33	24	21	20	
All-Sky Solar Radiation	RadAvg	544	890	1215	1463	1690	1879	1990	1660	1292	815	518	380	
	RadStd	62	88	109	144	138	145	70	120	90	108	48	45	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page